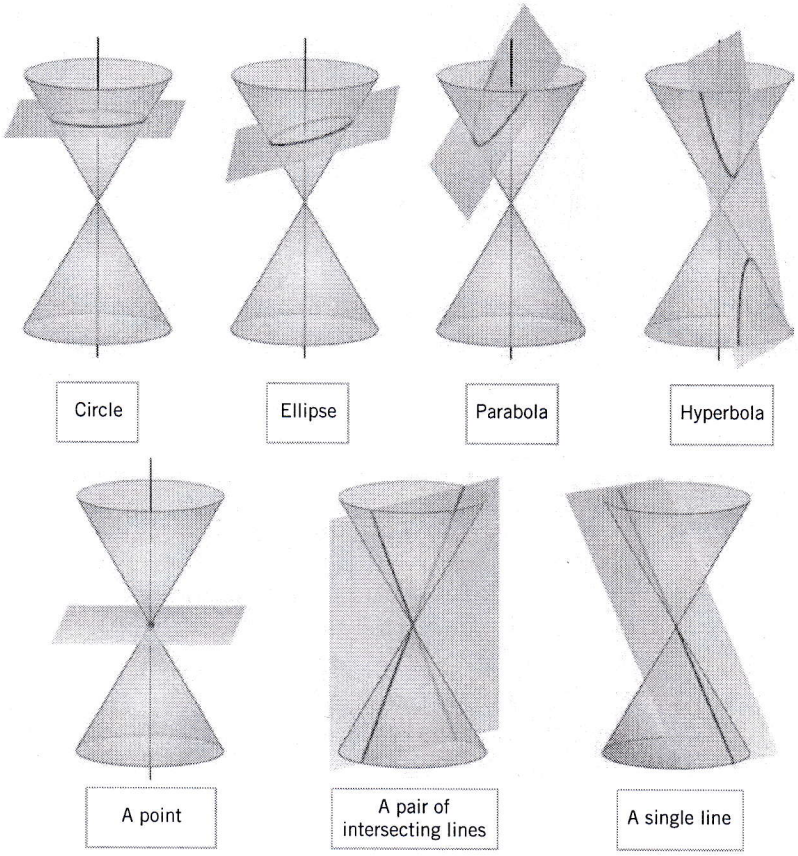
Week 10 Differential Calculus & Coordinate Geometry

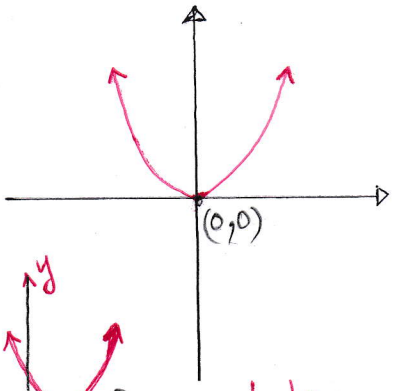
Conic section

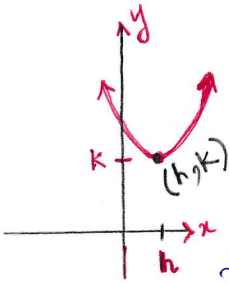
CONIC SECTIONS

Circles, ellipses, parabolas, and hyperbolas are called ***conic sections*** or ***conics*** because they can be obtained as intersections of a plane with a double-napped circular cone (Figure 10.4.1). If the plane passes through the vertex of the double-napped cone, then the intersection is a point, a pair of intersecting lines, or a single line. These are called ***degenerate conic sections***.



▲ **Figure 10.4.1**





parabola

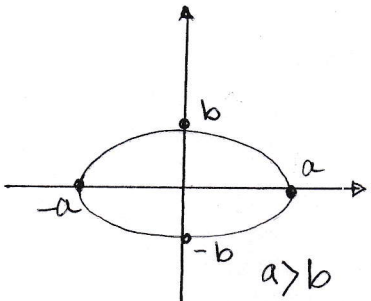
$y = x^2$

$x^2 = 4py$ → Standard form

Vertex: (0,0) (turning point)

$(x-h)^2 = \underbrace{4p}_{\text{constant}}(y-k)$; $p>0$ or $p<0$

Vertex: (h, k)

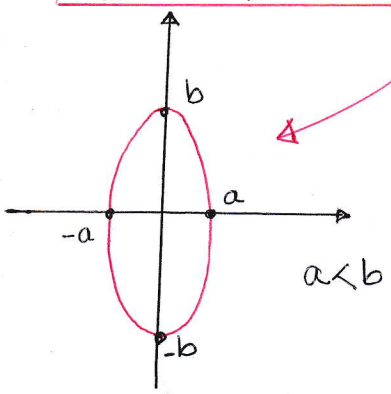


$a > b$

Ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ → Center: (0,0)

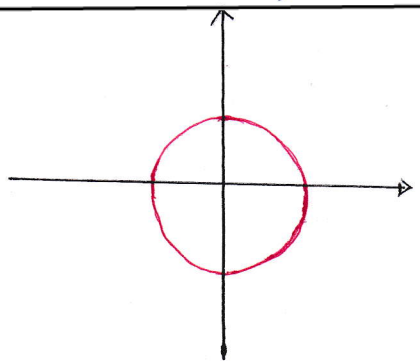
$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$



$a < b$

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ → Center: (h, k)

If it was not centered at the origin



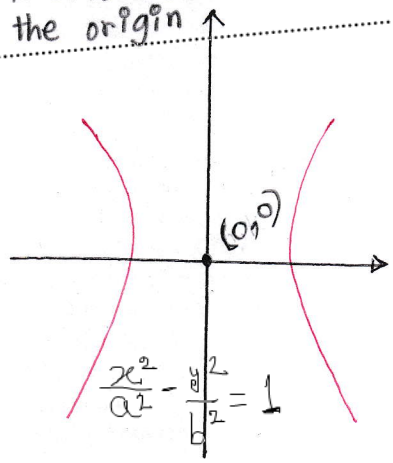
Circle

$x^2 + y^2 = r^2$

Center: (0,0)

$(x-h)^2 + (y-k)^2 = r^2$

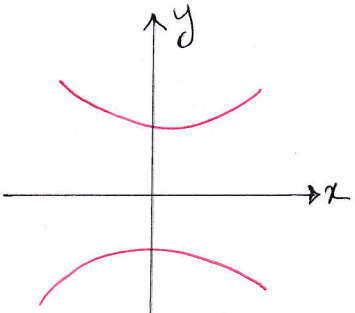
Center: (h, k)



$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Hyperbola

$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$



$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$

→ Center (h, k)

General Equation of Second Degree

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0 \quad \text{—— (1)}$$

or $$\boxed{ax^2+2hxy+by^2+2gx+2fy+c=0}$$ → (2) Relabeling the Constants

$$\begin{bmatrix} x & y & 1 \end{bmatrix}\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}=0$$

$A=a$; $B=2h$; $C=b$

$D=2g$; $E=2f$; $F=c$

We need to find the following Invariants

$$\Delta = abc+2fgh-af^2-bg^2-ch^2$$

discriminent of Eqn (1) → B^2-4AC, $(2h)^2-4ab$, $4h^2-4ab$, $4(h^2-ab)$, (h^2-ab) ↳ we can ignore the constant "4"

$$C = h^2-ab$$

$$I = a+b$$

Invariants are some unchanged constants which indicates the locus of conic.

Locus — a set of points, the position of which satisfy a set of algebric condition, such as circle, ellipse etc.

"Δ" identifies whether the conic is "Degenerate" (degenerate reduces quality of conic) or not (weak conic)

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = a(bc-f^2)-h(hc-fg)+g(hf-bg)$$

$$= abc-af^2-ch^2+fgh+fgh-bg^2$$

$$= abc+2fgh-af^2-bg^2-ch^2$$

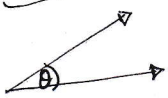
For degenerate case $\Delta = 0$; For proper conic $\Delta \neq 0$

Degenerate Case : $\Delta = 0$

If $h^2 - ab = 0$ then Parallel lines or straight line

If $a + b = 0$ then perpendicular line ⊥

If $\Delta = 0$ only but the above conditions do not fulfil then pair of st. lines.



Non-Degenerate Case : $\Delta \neq 0$

$h^2 - ab = 0 \longrightarrow$ Parabola

$h^2 - ab > 0 \longrightarrow$ hyperbola

$h^2 - ab < 0 \longrightarrow$ Ellipse

$h^2 - ab < 0$, $a = b$, $h = 0$ $\longrightarrow$ circle

What is represented by the following equation (s)?

(i) $4x^2 - 24xy - 6y^2 + 4x - 12y + 1 = 0$

a, $2h$, b, $2g$, $2f$, c

$2h = -24$
$h = -12$

$2g = 4$
$g = 2$

$2f = -12$
$f = -6$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= (4)(-6)(1) + 2(-6)(2)(-12) - 4(-6)^2 - (-6)(2)^2 - (1)(-12)^2$

$= 0$

Degenerate case

$h^2 - ab = (-12)^2 - (4)(-6) = 168 \neq 0$

$a + b = 4 + (-6) = -2 \neq 0$

↳ The eqn represents a pair of st. lines

(ii) $x^2 - 4xy + y^2 + 8x + 2y - 5 = 0$

$a = 1$, $2h = -4 \Rightarrow h = -2$; $b = 1$, $c = -5$; $2g = 8$, $g = 4$; $2f = 2$, $f = 1$

$h^2 - ab = (-2)^2 - (1)(1) = 3 > 0$

The eqn represents a hyperbola

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= -5 + (-16) - 1 - 16 - (-20)$

$= -18 \neq 0$

(iii) $8x^2 + 4xy + 5y^2 - 16x - 14y + 13 = 0$

$a = 8$, $b = 5$, $c = 13$; $2h = 4$, $h = 2$; $2g = -16$, $g = -8$; $2f = -14$, $f = -7$

$h^2 - ab = 2^2 - (8)(13) = -100 < 0$

The eqn represents an ellipse

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= 520 + 224 - 392 - 320 - 52$

$= -20 \neq 0$

(iv) $x^2 - 6xy + 9y^2 + 4x + 8y + 15 = 0$

$a = 1$, $b = 9$, $c = 15$; $2h = -6$, $h = -3$; $2g = 4$, $g = 2$; $2f = 8$, $f = 4$

$h^2 - ab = (-3)^2 - (1)(9) = 0$

$\therefore h^2 - ab = 0$, It could be degenerate or parabola

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= (1)(9)(15) + 2(4)(2)(-3) - (1)(4)^2 - (9)(2)^2 - (15)(-3)^2$

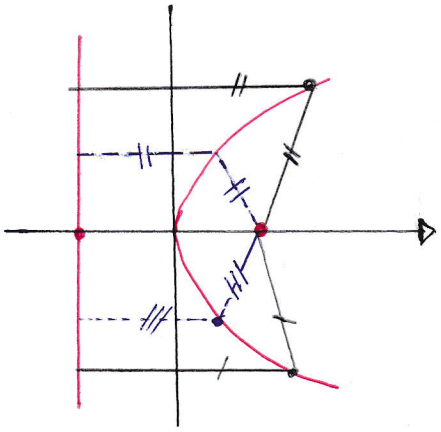
$= -100$

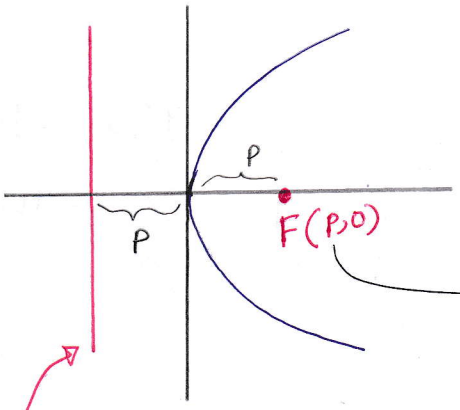
$\therefore \Delta \neq 0$

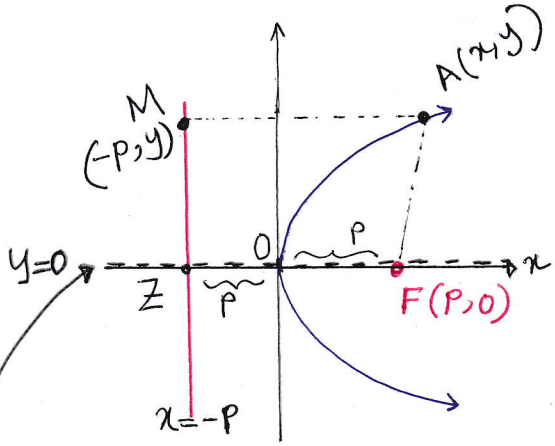
Hence the eqn represents a parabola.

Parabola

Set of points in a plane that are equidistant from a fixed point F (focus) and a fixed line (directrix)







$OZ = OF$

Vertex, $O = (0,0)$

Focus, $F = (p,0)$

Eqn of directrix, MZ: $x = -p$

$FA = AM$

$$\sqrt{(x-p)^2+(y-0)^2} = \sqrt{(x-(-p))^2+(y-y)^2}$$

$$\sqrt{(x-p)^2+y^2} = \sqrt{(x+p)^2}$$

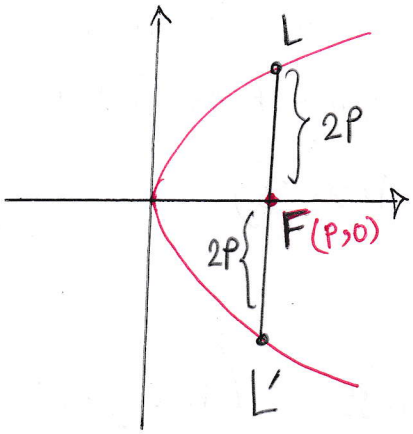
$$(x-p)^2+y^2 = (x+p)^2$$ } square both sides

$$x^2-2xp+p^2+y^2 = x^2+2xp+p^2$$

$$y^2 = 4px$$

$(y-k)^2 = 4p(x-h)$ → if the Vertex shifts from the origin (0,0)

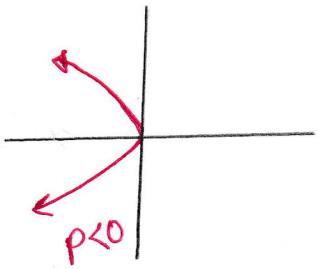
Axis of symmetry: the line that divides the parabola into two equal parts: $y=0$



LL' : Latus Rectum

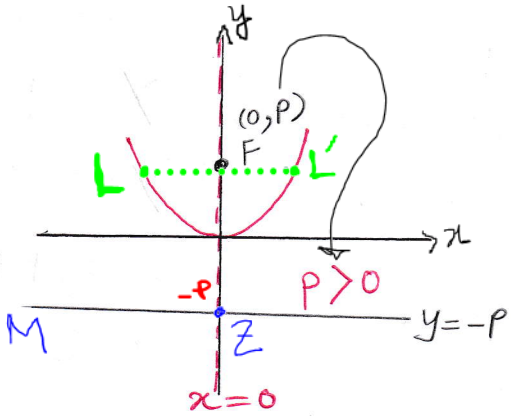
$\therefore LL' = 4p$

$p > 0$ or $p < 0$



Eqn of Latus Rectum: $x = p$

Length of Latus Rectum: $LL' = |4p|$ → Length is always +ve



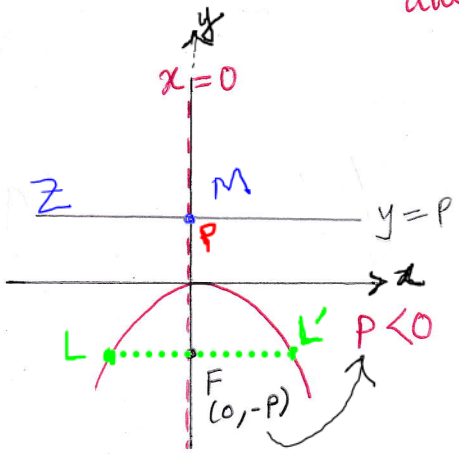
Vertex: (0,0)

Focus: (0, p)

(directrix) mz : $y = -p$

axis of symmetry: $x = 0$

Latus Rectum, LL': $|4p|$



Vertex: (0,0)

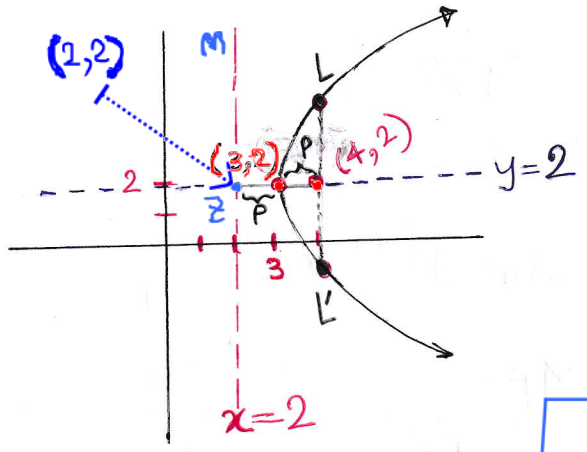
Focus: (0,-p)

mz : $y = p$

axis of symmetry: $x = 0$

Latus Rectum, $LL' = |4p|$

Find the vertex, focus, equation of directrix, equation of axis of symmetry, length of Latus rectum of the parabola: $(y-2)^2 = 4(x-3)$



$(y-2)^2 = 4(x-3)$

$Y^2 = 4pX$

$(h,k) = (3,2)$

$4p = 4$

$p = 1$

$Y=0 \Rightarrow y-2=0$

$y=2$

$X=0 \Rightarrow x-3=0$

$x=3$

Vertex: $(3,2)$

Focus: $(p,0)$

$= (4,2)$

$X = p$ $\quad Y=0$

$x-3=p$ $\quad y-2=0$

$x-3=1$ $\quad y=2$

$x=4$

Vertex: $(3,2)$

Focus: $(4,2)$

Eqn of directrix: $x=2$

Eqn of axis:

$Y=0$

$y-2=0$

$y=2$

Eqn of directrix: $X=-p$

$\Rightarrow X+p=0$

$\Rightarrow (x-3)+p=0$

$\Rightarrow x=2$

Length of Latus Rectum: $|4p|$

(LL') $= |4(1)|$

$= 4$

Find the vertex, focus, directrix, axis and length of latus rectum for the parabola, $y^2-2y-4x-3=0$

$y^2-2y-4x-3=0$ ⟶ General form

$y^2-2\cdot y\cdot 1+1^2-1^2-4x-3=0$ completing square

$(y-1)^2-4x-4=0$

$(y-1)^2=4(x+1)$ ⟶ standard form

$(y-1)^2=4p(x+1)$

$Y^2=4pX$

$4p=4$

$\therefore p=1$

(i) Vertex: $(-1,1)$

(ii) Focus: $(p,0)=(0,1)$

$X=p$

$x+1=1$

$x=0$

$Y=0$

$y-1=0$

$y=1$

(iii) Eqn of directrix: $x=-2$ (MZ)

$X=-p$

$x+1=-1$

$x=-2$

(iv) Eqn of axis: $y=1$

$Y=0$

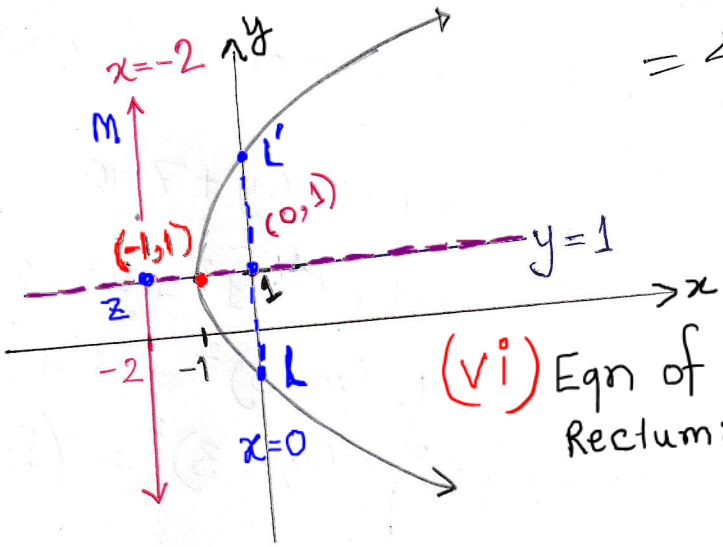
$y-1=0$

$y=1$

(v) Length of Latus Rectum: $|4p|$

$=|4(1)|$

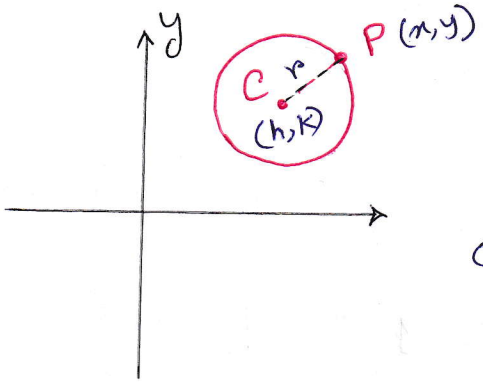
$=4$



(vi) Eqn of Latus Rectum: $x=0$

Circle

Set of points in a plane that are equidistant from a fixed point C (center).



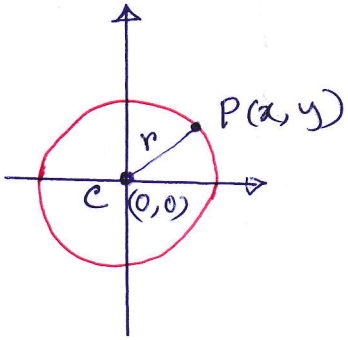
$$|PC| = \sqrt{(x-h)^2+(y-k)^2} = r$$

$$\Rightarrow \boxed{(x-h)^2+(y-k)^2 = r^2}$$

Square both sides

center: (h,k)

Standard eqn of the circle.

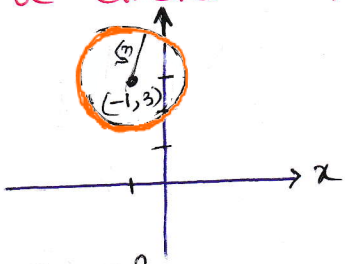


$$|PC| = \sqrt{(x-0)^2+(y-0)^2} = r$$

$$\Rightarrow x^2+y^2 = r^2$$

Center: (0,0)

Question: Sketch the graph with the equation $x^2+y^2+2x-6y+7=0$ by first showing that it represents a circle and then finding its center and radius.



$$x^2+2x+y^2-6y+7=0$$

$$\Rightarrow x^2+2\cdot x\cdot 1+1^2-1^2+y^2-2\cdot y\cdot 3+3^2-3^2+7=0$$

$$\Rightarrow (x+1)^2+(y-3)^2 = 1^2+3^2-7 = 3$$

$$\Rightarrow (x+1)^2+(y-3)^2 = (\sqrt{3})^2$$

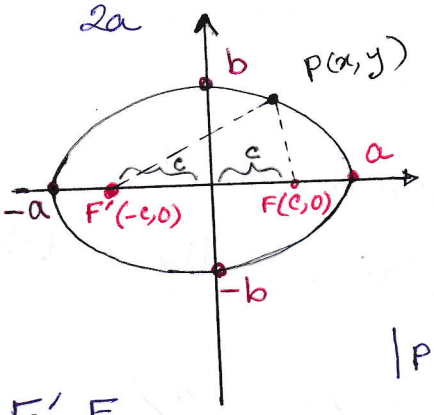
Center: (-1, 3)

radius: $\sqrt{3}$

Ellipse

An ellipse is the set of all points in the plane the sum of whose distances from two fixed points (the foci) is a constant.

foci → plural of focus



$a > b$

F', F → foci

$$|PF'| + |PF| = 2a$$

$$\Rightarrow \sqrt{(x+c)^2+y^2} + \sqrt{(x-c)^2+y^2} = 2a$$

$$\Rightarrow \sqrt{(x-c)^2+y^2} = 2a - \sqrt{(x+c)^2+y^2}$$

$$\Rightarrow x^2 - 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x+c)^2+y^2} + x^2 + 2cx + c^2 + y^2$$

(sq both sides)

$$\Rightarrow a\sqrt{(x+c)^2+y^2} = a^2 + cx$$

$$\Rightarrow a^2(x^2+2cx+c^2+y^2) = a^4 + 2a^2cx + c^2x^2$$

(sq both sides)

$$\Rightarrow a^2x^2 + 2a^2cx + a^2c^2 + a^2y^2 = a^4 + 2a^2cx + c^2x^2$$

$$\Rightarrow a^2x^2 - c^2x^2 + a^2y^2 = a^4 - a^2c^2$$

$$\Rightarrow (a^2-c^2)x^2 + a^2y^2 = a^2(a^2-c^2)$$

$$\Rightarrow b^2x^2 + a^2y^2 = a^2b^2$$

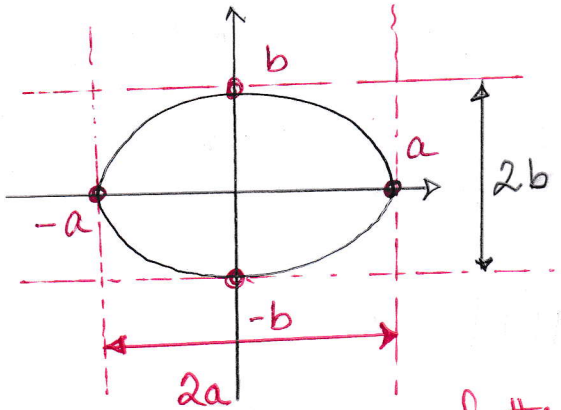
Redefine the constant $a^2 - c^2 = b^2$

$$\Rightarrow \frac{b^2x^2}{a^2b^2} + \frac{a^2y^2}{a^2b^2} = \frac{a^2b^2}{a^2b^2} \quad (\div \text{ by } a^2b^2)$$

$$\Rightarrow \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \rightarrow \text{Eqn of Ellipse}$$

Center $(0,0)$

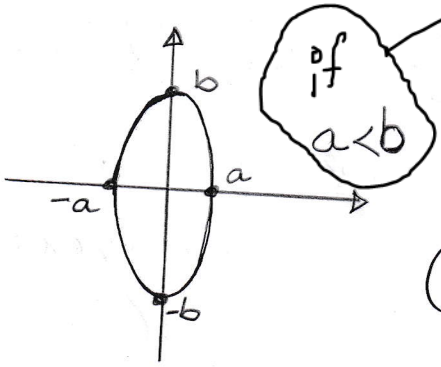
$a > b$



$2b \rightarrow$ distance of minor axis $\because a > b$

(Horizontally stretched)

$2a \rightarrow$ distance of the major axis. $\because a > b$

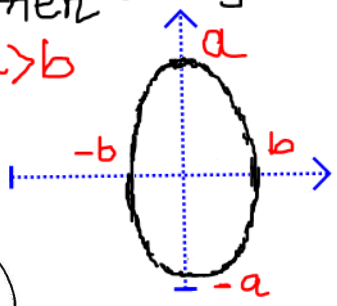


if $a < b$

then change it

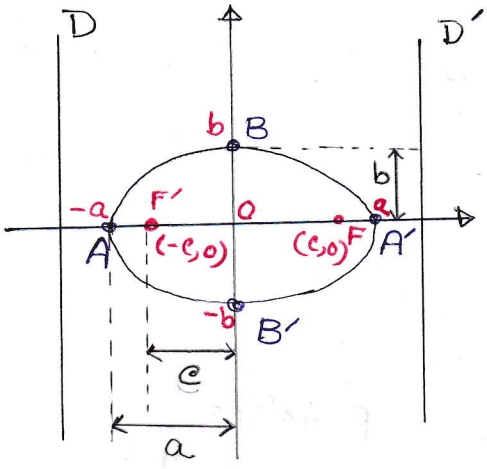
$a > b$

$$\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$$



(Vertically stretched)

Ratio between the length of the focus and the length of the semi major axis (one of the vertices)



Eccentricity, $e = \frac{c}{a}$

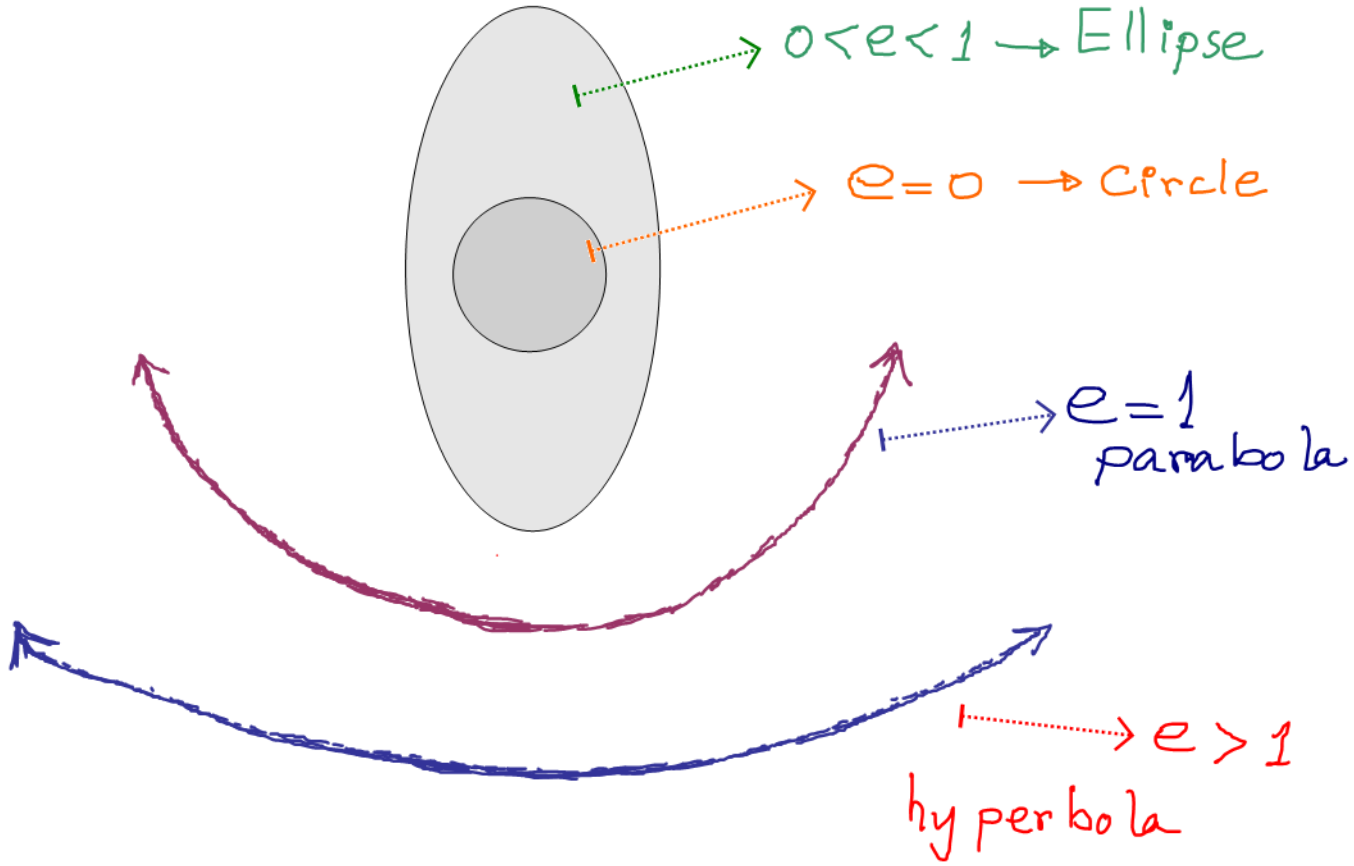
↓

It is a constant

{ It defines how circular or uncircular the shape of the conic is }

$0 < e < 1$

$\boxed{c < a}$



$$e = \frac{c}{a}$$

$a^2 - c^2 = b^2$ ∵ we redefined this constant

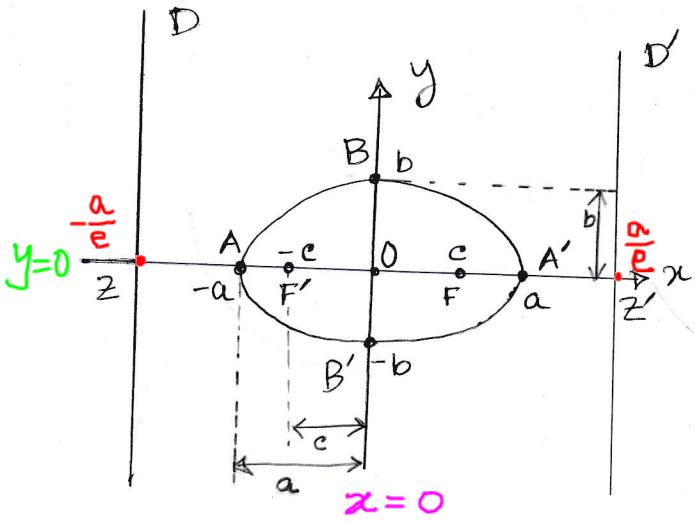
$$e = \frac{\sqrt{a^2-b^2}}{a}$$

$$\Rightarrow c^2 = a^2 - b^2$$

$$\Rightarrow c = \sqrt{a^2-b^2}$$

$$= \sqrt{\frac{a^2-b^2}{a^2}}$$

$$= \sqrt{1 - \frac{b^2}{a^2}}$$



Centre, O : (0,0)

Foci, $(\pm c, 0)$ or $(\pm ae, 0)$

↳ F' & F

$$\begin{cases} e = \frac{c}{a} \\ c = ae \end{cases}$$

Vertices $(\pm a, 0)$ → A & A'

Covertices $(0, \pm b)$ → B & B'

Equation of major axis: $\boxed{y=0}$ → AA' (the line)

Equation of minor axis: $\boxed{x=0}$ → BB' (the line)

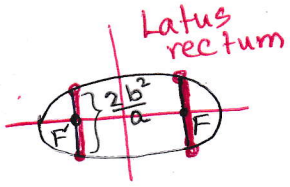
Length of major axis: $2a$ (AA')

Length of minor axis: $2b$ (BB')

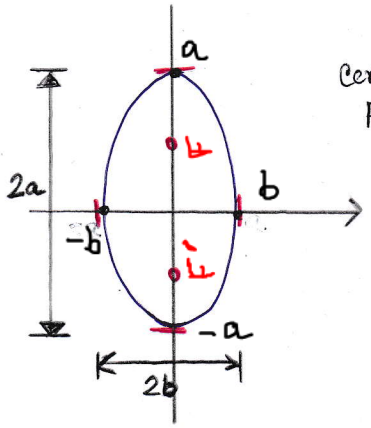
Equation of directrices: DZ & D'Z'

$$\boxed{x = \pm \frac{a}{e}}$$

Equation of latus rectum: $\begin{cases} x = \pm c \\ x = \pm ae \\ \because c = ae \end{cases}$



$$\boxed{\text{Length of Latus rectum: } \frac{2b^2}{a}}$$



center: (0,0)

Foci : $(0, \pm ae)$

Vertices: $(0, \pm a)$ → major axis

Covertices: $(\pm b, 0)$ → minor axis

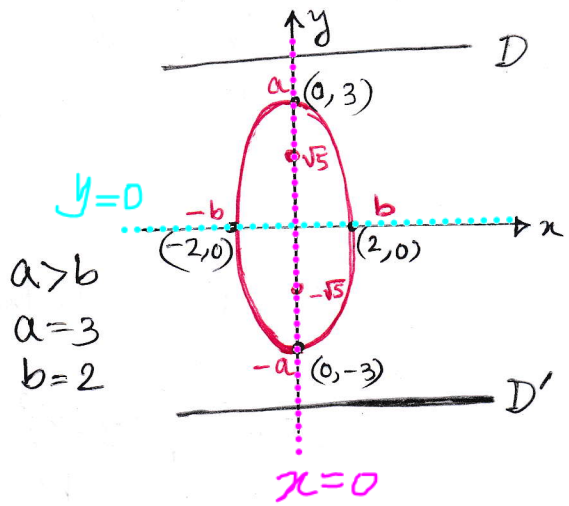
Example: Find all the characteristics of the graph with equation: $9x^2+4y^2=36$

$\frac{y^2}{a^2}+\frac{x^2}{b^2}=1$

$\frac{x^2}{4}+\frac{y^2}{9}=1 \quad (\div \text{by } 36)$

$\Rightarrow \frac{x^2}{2^2}+\frac{y^2}{3^2}=1$

$\Rightarrow \frac{y^2}{3^2}+\frac{x^2}{2^2}=1$



$a>b$
$a=3$
$b=2$

Center: $(0,0)$

Vertices: $(0,\pm a)=(0,\pm 3)$ → major axis

Covertices: $(\pm b,0)=(\pm 2,0)$ → minor axis

$e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{4}{9}}=\sqrt{\frac{9-4}{9}}=\frac{\sqrt{5}}{3}$

Foci: $(0,\pm ae)=(0,\pm\sqrt{5})$ $\quad ae=3\cdot\frac{\sqrt{5}}{3}=\sqrt{5}$

Equation of major axis: $x=0$

Eqn of minor axis: $y=0$

Length of major axis: $2a=2(3)=6$

Length of minor axis: $2b=4$

Equation of directrices: $y=\pm\frac{a}{e}=\pm\frac{9}{\sqrt{5}}$ $\quad \begin{cases} a=3 \\ e=\frac{\sqrt{5}}{3} \\ b=2 \end{cases}$

Eqn of latus rectum: $y=\pm ae=\pm\sqrt{5}$

Length of latus rectum: $\frac{2b^2}{a}=\frac{8}{3}$

Compare

$$\frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1$$

$$\frac{Y^2}{a^2}+\frac{X^2}{b^2}=1$$

if center is not at the origin;
Center: (h,k)

Standard forms of the eqn of an ellipse with center: (h,k)

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

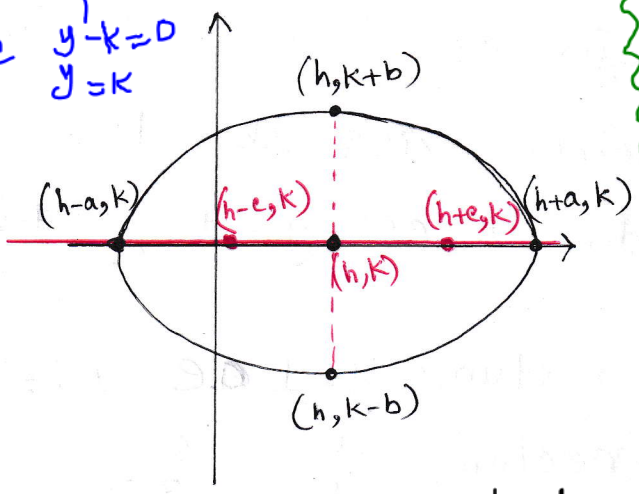
- $a>b$
- Length of major axis $=2a$
- " " minor " $=2b$
- Coordinates of the vertices: $(h\pm a,k)$
- Coordinates of the covertices: $(h,k\pm b)$
- Coordinate of the foci: $(h\pm c,k)$

$(\pm a,0)$
$X=\pm a$ $\quad Y=0$
$x-h=\pm a$ $\quad y-k=0$
$x=h\pm a$ $\quad y=k$

$(0,\pm b)$
$X=0$ $\quad Y=\pm b$
$x-h=0$ $\quad y-k=\pm b$
$x=h$ $\quad y=k\pm b$

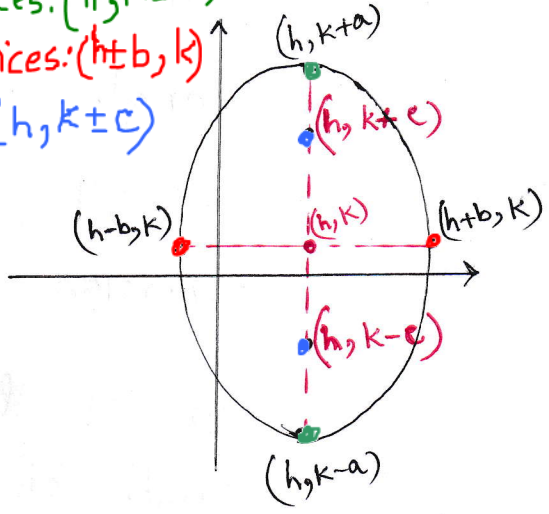
$(\pm c,0)$
$X=\pm c$
$x-h=\pm c$
$x=h\pm c$

$Y=0$
$y-k=0$
$y=k$



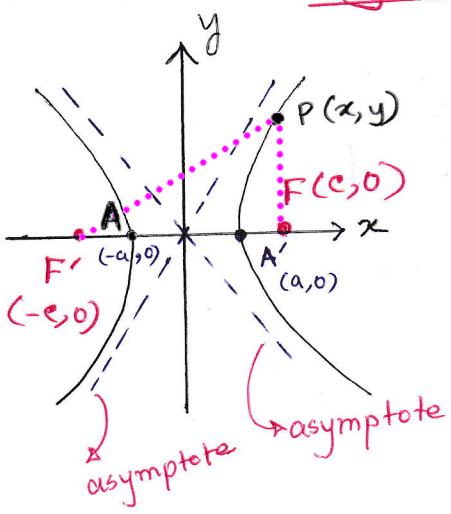
Horizontally stretched

Vertices: $(h,k\pm a)$
Covertices: $(h\pm b,k)$
foci: $(h,k\pm c)$



Vertically stretched

Hyperbola



set of all points in a plane the difference of whose distances from two fixed points (the foci) is a constant. ($2a$)

F & F'

$|PF| - |PF'| = \pm 2a$

$PF > PF' = 2a$

$PF < PF' = -2a$

$$\Rightarrow \sqrt{(x-c)^2+y^2} - \sqrt{(x+c)^2+y^2} = \pm 2a$$

$$\Rightarrow \sqrt{(x-c)^2+y^2} = \pm 2a + \sqrt{(x+c)^2+y^2}$$

$$\Rightarrow x^2 - 2cx + c^2 + y^2 = 4a^2 \pm 4a\sqrt{(x+c)^2+y^2} + x^2 + 2cx + c^2 + y^2$$

$$\Rightarrow -4cx = 4a^2 \pm 4a\sqrt{(x+c)^2+y^2}$$

$$\Rightarrow \pm a\sqrt{(x+c)^2+y^2} = a^2 + cx$$

$$\Rightarrow a^2(x^2+2cx+c^2+y^2) = a^4 + 2a^2cx + c^2x^2$$

$$\Rightarrow a^2x^2 + 2a^2cx + a^2c^2 + y^2a^2 = a^4 + 2a^2cx + c^2x^2$$

$$\Rightarrow a^2x^2 + a^2c^2 + y^2a^2 = a^4 + c^2x^2$$

$$\Rightarrow a^2x^2 + a^2(a^2+b^2) + a^2y^2 = a^4 + (a^2+b^2)x^2$$

$$\Rightarrow a^2x^2 + a^4 + a^2b^2 + a^2y^2 = a^4 + a^2x^2 + b^2x^2$$

$$\Rightarrow a^2b^2 = b^2x^2 - a^2y^2$$

$$\Rightarrow \boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$ ($\div$ by a^2b^2)

→ standard form

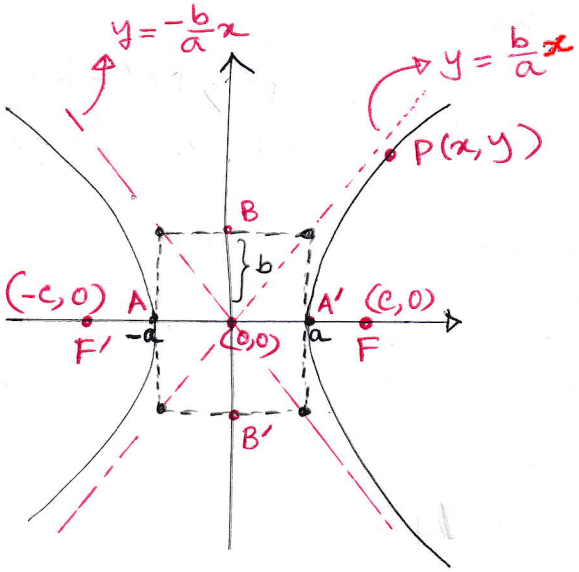
Redefine the constant

$a^2 - c^2 = b^2$

$\Rightarrow c^2 = a^2 + b^2$

Center: $(0,0) \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Center: $(h,k) \Rightarrow \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$



Centers: C (0,0)

Vertices: A & A'

$\Rightarrow (\pm a, 0)$

Foci: F & F'

$\Rightarrow (\pm c, 0)$ or $(\pm ae, 0)$

$e > 1$

See graph at pg 13

Eccentricity: $e = \sqrt{1 + \frac{b^2}{a^2}}$: $e = \frac{c}{a}$, $e > 1$

↑ The sign is opposite to ellipse

Eqn of Asymptoses: $y = \pm \left(\frac{b}{a}\right)x$

Length of transverse axis: $2a$ (AA')

Length of conjugate axis: $2b$ (BB')

Eqn of directrices: $x = \pm \frac{a}{e}$ (D & D')

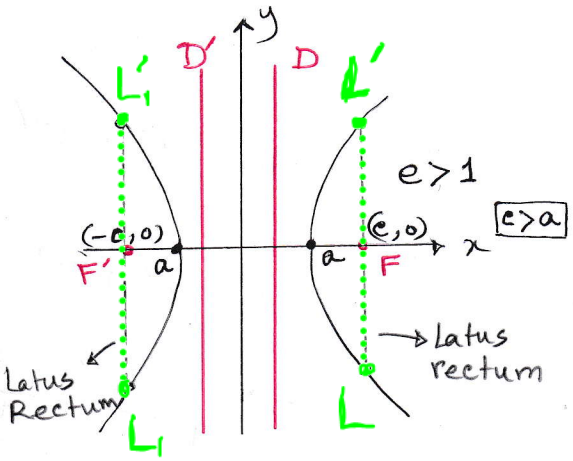
same as ellipse → Length of Latus rectum: $\frac{2b^2}{a}$

Eqn of Latus rectum:

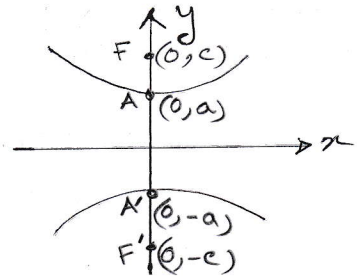
$x = -c$, $x = c$

$= -ae$ $= ae$

Foci



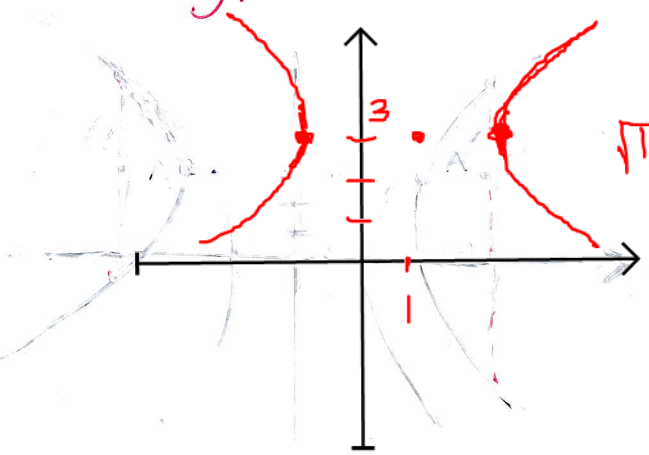
If we have: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$



Vertices: $(0, \pm a)$

Foci: $(0, \pm c)$ or $(0, \pm ae)$

Example: Find all the characteristics of the hyperbola with equation $2(x-1)^2 - 3(y-3)^2 = 30$



$2(x-1)^2 - 3(y-3)^2 = 30$

$\frac{(x-1)^2}{15} - \frac{(y-3)^2}{10} = 1$

$\sqrt{15} > \sqrt{10}$

$a = \sqrt{15}$, $b = \sqrt{10}$

Center, $C = (1,3)$

Vertices $= (1 \pm \sqrt{15}, 3)$

$\frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1$ Vertices: $(\pm a, 0)$

$X = \pm a$ $Y = 0$

$\Rightarrow x - 1 = \pm a$ $\Rightarrow y - 3 = 0$

$\Rightarrow x = 1 \pm a$ $\Rightarrow y = 3$

$x = 1 \pm \sqrt{15}$

$e = \sqrt{1 + \frac{b^2}{a^2}}$

$= \sqrt{1 + \frac{10}{15}}$

$e = \sqrt{\frac{25}{15}} = \frac{\sqrt{5}}{\sqrt{3}} = \frac{\sqrt{15}}{3}$

Foci: $(\pm ae, 0)$

$X \quad Y$

$X = \pm ae$

$x - 1 = \pm \sqrt{15}\left(\frac{\sqrt{15}}{3}\right) = \pm \frac{15}{3} = \pm 5$

$x = 1 \pm 5 = -4, 6$

$Y = 0$

$y - 3 = 0$

$y = 3$

Foci: (6, 3) and (−4, 3)

Eqn of directrices: $X = \pm \frac{a}{e}$

$$x-1 = \pm \frac{\sqrt{15}}{\sqrt{15}/3} = \pm 3$$

$$x = 1 \pm 3$$

$$x = 4, \quad x = -2$$

Eqn of asymptotes: $Y = \pm \frac{b}{a} X$

$$y - 3 = \pm \frac{\sqrt{10}}{\sqrt{15}} (x-1)$$

$$y = 3 \pm \frac{\sqrt{2}}{\sqrt{3}} (x-1)$$

$$= 3 \pm \frac{\sqrt{6}}{3} (x-1)$$

$$y = \frac{\sqrt{6}}{3}(x-1) + 3 \; ; \quad y = -\frac{\sqrt{6}}{3}(x-1) + 3$$

Eqn of latus rectum:

$\underbrace{x = 6 \text{ and } x = -4}_{\text{Foci}}$

Length of Latus rectrum: $\frac{2b^2}{a} = \frac{2(\sqrt{10})^2}{\sqrt{15}}$

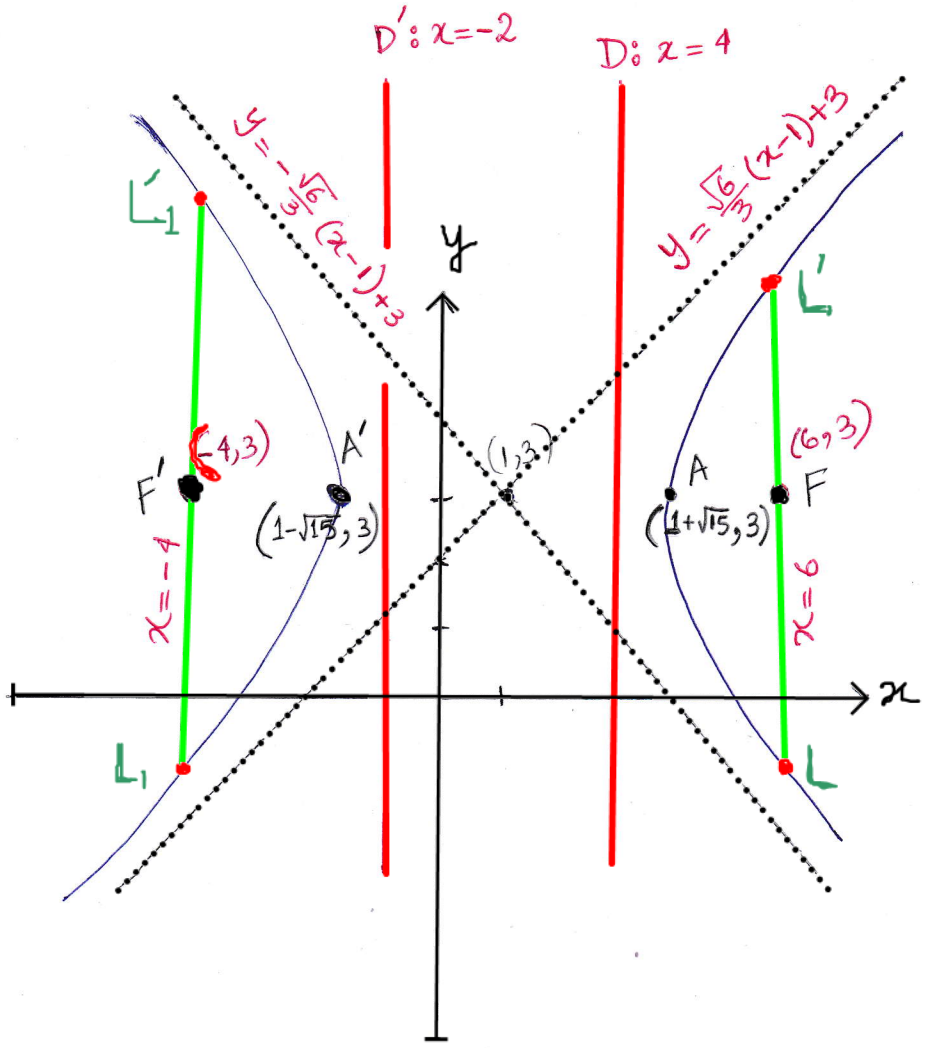
$$= \frac{20}{\sqrt{15}}$$

$$= \frac{20\sqrt{15}}{15}$$

$$= \frac{4\sqrt{15}}{3}$$

Length of tranverse axis : $2a = 2\sqrt{15}$

" " conjugate axis: $2b = 2\sqrt{10}$


D′: x=-2
D: x=4
y=-√6/3 (x-1)+3
y=√6/3 (x-1)+3
L′1
L′
y
x
(-4,3)
F′
A′
(1-√15, 3)
(1,3)
A
(1+√15, 3)
F
(6,3)
x=-4
x=6
L1
L